ECE 269 Mini Project 1: Finding Sparse Solutions via Orthogonal Matching Pursuit (OMP)

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1 Methods

1.1 Orthogonal Matching Pursuit (OMP)

The task of this project is to recover a sparse signal $x \in \mathbb{R}^N$ from compressive measurements $y \in \mathbb{R}^M$ with $M < N$. It assumes the linear measurement model is $y = Ax + n$, where $A = [a_1, \ldots, a_N] \in \mathbb{R}^{M \times N}$ is the measurement matrix and n denotes measurement noise. To solve this problem, OMP [1] is utilized: it estimates the support of x and reconstructs x from y by iteratively selecting the column of A most correlated with the current residual, followed by a least-squares refit on the selected support. A rough process of OMP is shown below.

Algorithm 1 Orthogonal Matching Pursuit (OMP)

Require: $A \in \mathbb{R}^{M \times N}$ (columns $\{a_i\}$), $y \in \mathbb{R}^M$, stopping parameters (s, $\|n\|_2$, or $s_{\max}$)
Ensure: estimate $x \in \mathbb{R}^N$ and support S

```
S ← ∅, x ← 0, r ← y
for k = 1, 2, ... do
    c ← A^⊤ r
    i_k ← arg max_{i ∉ S} |c_i|
    S ← S ∪ {i_k}
    x_S ← arg min_z ||y − A_S z||_2^2
    x_{S^c} ← 0                          ▷ all unselected entries are zero
    r ← y − Ax
    if stopping rule satisfied then
        break
    end if
end for
return x, S
```

Stopping rules used in this project.

- **Noiseless:** stop when $\|r\|_2$ is numerically close to 0 (e.g., $\|r\|_2/\|y\|_2 < \varepsilon$) or after $s_{\max}$ iterations.
- **Noisy (known s):** stop after selecting s atoms (i.e., $|S| = s$).
- **Noisy (known $\|n\|_2$):** stop when $\|r\|_2 \leq \|n\|_2$.

After the least-squares refit on S, the residual corresponds to the component of y not explained by span(A_S) (i.e., the LS residual is orthogonal to span(A_S)).

1.2 Least Squares (LS)

- Solve $x_{\text{LS}} = \arg\min_x \|Ax - y\|_2^2$ with `np.linalg.lstsq`.

- For $M < N$, LS returns a minimum-norm solution and typically does not promote sparsity.

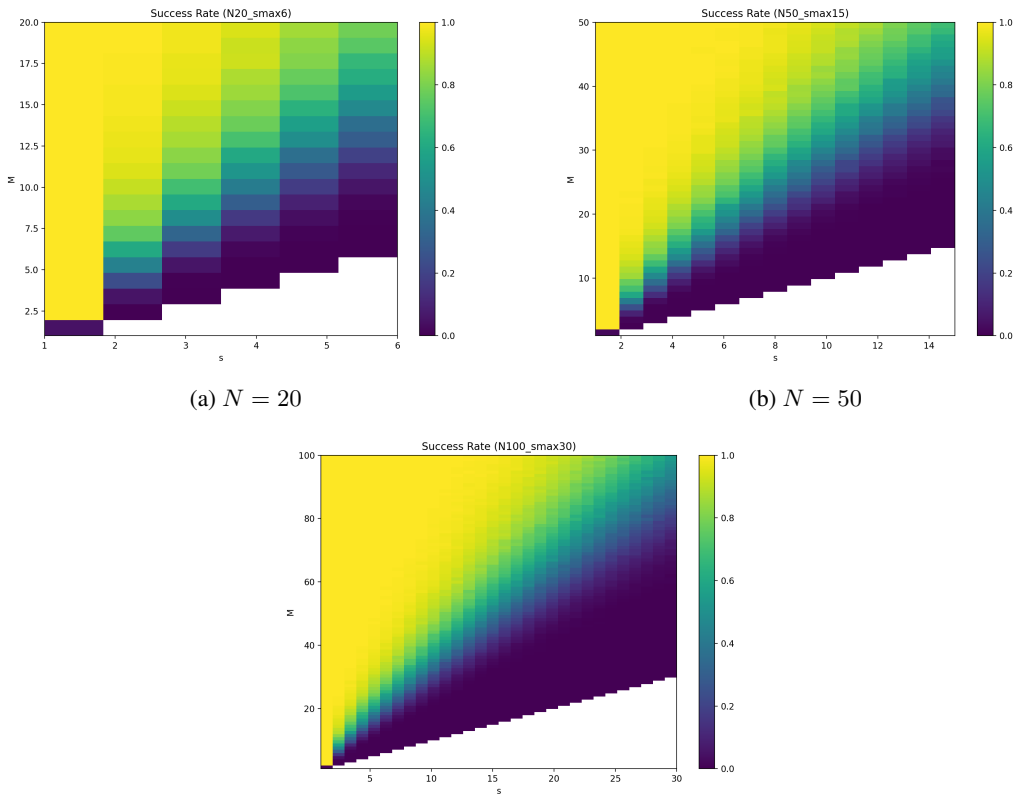
1.3 Implementation Details

- **Column normalization (if used):** normalize each column of A to unit ℓ_2 norm so that $|a_i^\top r|$ reflects correlation rather than column magnitude.
- **Error metric:** Normalized Error $\frac{\|x_{\text{est}} - x\|_2}{\|x\|_2}$, and success if $< 10^{-3}$ when required.

2 Noiseless case ($n = 0$)

The performance of OMP in the noiseless setting is evaluated by the probability of Exact Support Recovery (i.e., the fraction of trials where $\hat{S} = S$ over 2000 random realizations of A) and the average Normalized Error $\|x_{\text{est}} - x\|_2 / \|x\|_2$ as functions of (M, s) for $N \in \{20, 50, 100\}$. Since the task focuses on the sparse regime $s \ll N$ and the runtime grows noticeably when s becomes large, I only sweep sparsity up to $s \leq \lfloor 0.3M \rfloor$. Regions outside this range are left blank in the heatmaps.

From Fig. 1, there is a clear transition boundary: when M is sufficiently large relative to s, OMP recovers the exact support with probability close to 1; below the boundary, the success probability drops rapidly toward 0. The boundary is monotone in s : larger sparsity requires more measurements. As N increases from 20 to 100, the transition shifts slightly upward, meaning the same (M, s) becomes marginally harder at larger ambient dimension.



(a) $N = 20$

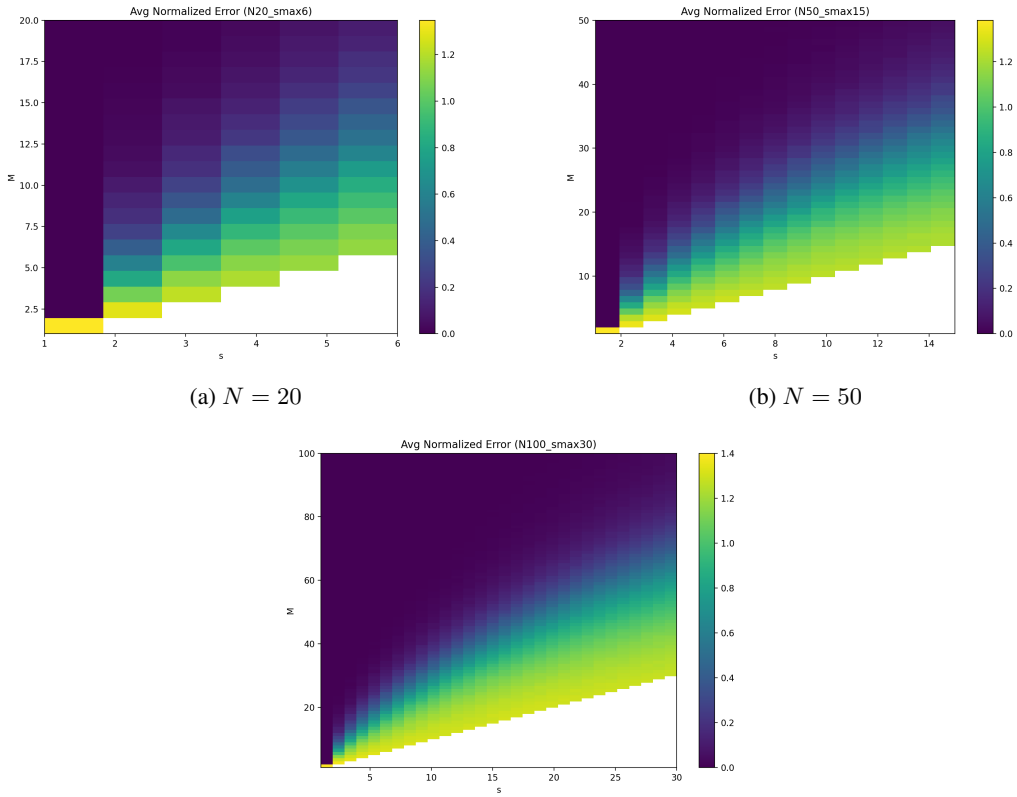
(b) $N = 50$

(c) $N = 100$

Figure 1: Noiseless phase transition (Exact Support Recovery). Each heatmap shows the empirical success probability $\mathbb{P}(\hat{S} = S)$ over 2000 trials versus sparsity s (x-axis) and measurements M (y-axis).

Fig. 2 is consistent with the success-rate plots shown in Fig. 1. In the region where exact support recovery is likely, the normalized error is near zero. Near the transition boundary the error increases

more gradually than the success probability, because exact support recovery is a strict criterion: OMP may fail to identify the exact support while still producing a partially accurate approximation, leading to moderate error before entering the clear failure region.



(a) $N = 20$

(b) $N = 50$

(c) $N = 100$

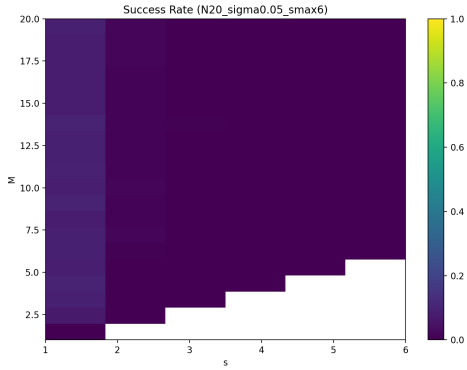
Figure 2: Noiseless phase transition (Average Normalized Error). Each heatmap shows the average $\|x_{\text{est}} - x\|_2 / \|x\|_2$ over 2000 trials versus (M, s).

3 Noisy case ($n \neq 0$)

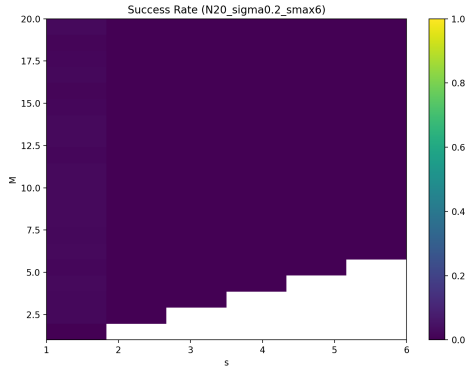
In the noisy setting, the measurement model is $y = Ax + n$, where the noise is generated as i.i.d. Gaussian entries $n_i \sim \mathcal{N}(0, \sigma^2)$. I test two noise levels $\sigma \in \{0.05, 0.2\}$ and report phase-transition heatmaps over (M, s) for $N \in \{20, 50, 100\}$. Success is defined by Normalized Error $< 10^{-3}$. For the same reason in the noiseless case, I only take $s \leq \lfloor 0.3M \rfloor$.

3.1 Known sparsity s: noisy phase transition

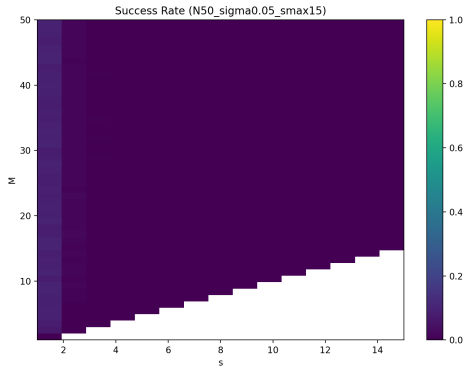
When the sparsity level s is known, OMP stops after selecting s atoms. Fig. 3 shows success-rate heatmaps for both noise levels. For particularly small s, the lower noise case ($\sigma = 0.05$) achieves slightly higher success probability than $\sigma = 0.2$, since noise can perturb the correlation scores used for atom selection. As s increases, recovery becomes nearly impossible as the success rate is close to zero and there is nearly no difference between $\sigma = 0.05$ and $\sigma = 0.2$.



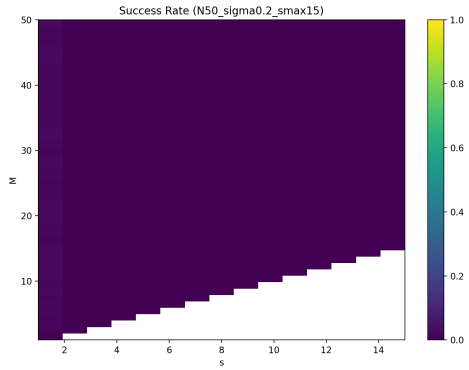
(a) $N = 20,\ \sigma = 0.05$



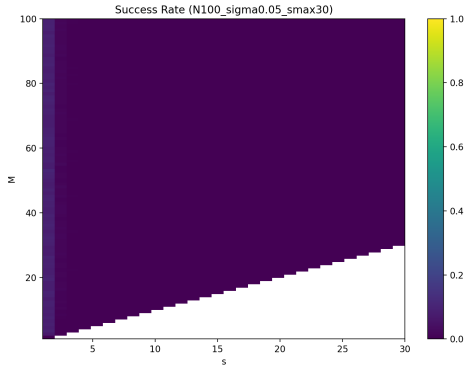
(b) $N = 20,\ \sigma = 0.2$



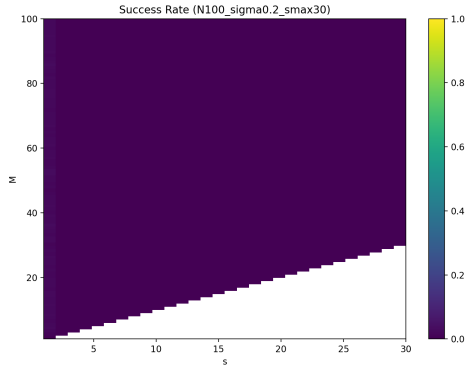
(c) $N = 50,\ \sigma = 0.05$



(d) $N = 50,\ \sigma = 0.2$

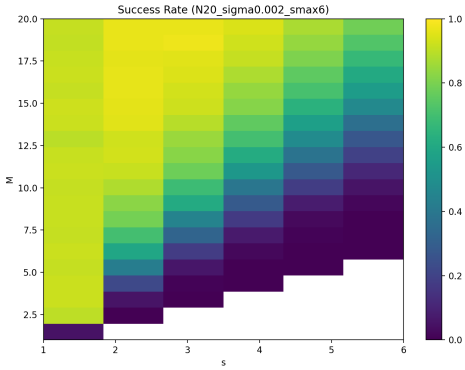


(e) $N = 100,\ \sigma = 0.05$

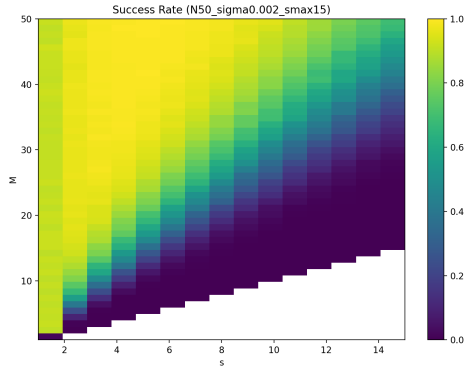


(f) $N = 100,\ \sigma = 0.2$

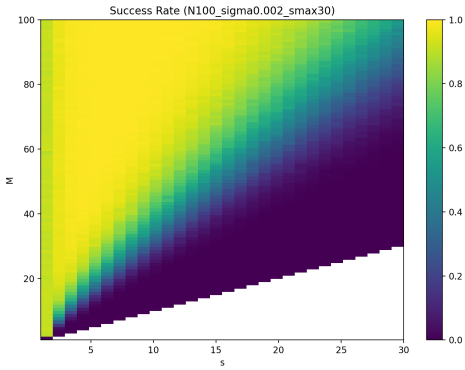
Figure 3: Noisy phase transition (known s): success rate for $N \in \{20, 50, 100\}$ and $\sigma \in \{0.05, 0.2\}$. Success is defined as Normalized Error $< 10^{-3}$.



(a) $N = 20,\ \sigma = 0.002$



(b) $N = 50,\ \sigma = 0.002$



(c) $N = 100,\ \sigma = 0.002$

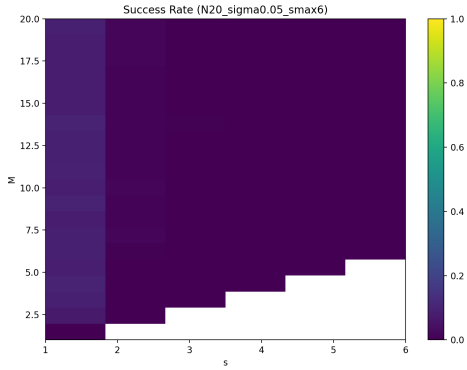
Figure 4: Additional noisy phase transition results under a smaller noise level $\sigma = 0.002$.

Since the success rate bias between $\sigma = 0.05$ and $\sigma = 0.2$ is extremely small, I conducted additional experiments with $\sigma = 0.002$ to compare with $\sigma = 0.2$. Still, I only took $s \leq \lfloor 0.3M \rfloor$ and other settings are still the same. The result is shown in Fig. 4.

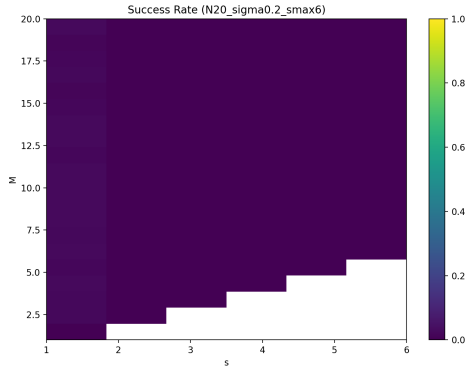
With $\sigma = 0.002$, the success-rate heatmaps show a much clearer phase-transition boundary compared to larger noise levels. The high-success region expands, and the transition becomes sharper, which is consistent with the intuition that smaller noise makes OMP support selection more reliable.

3.2 Unknown s but known $\|n\|_2$: stopping by residual norm

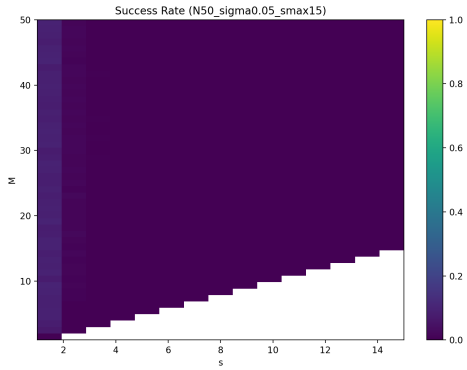
In this setting, s is unknown and OMP stops when the residual norm satisfies $\|y - Ax^{(k)}\|_2 \leq \|n\|_2$. Fig. 5 shows success-rate heatmaps. Compared with the known-s case, this stopping rule can be more conservative: stopping early may avoid overfitting noise but can also terminate before selecting enough atoms, which lowers the probability of meeting the strict success criterion. But here I can not see the difference. The impact of noise level is still clear when s is small: $\sigma = 0.05$ generally yields better performance than $\sigma = 0.2$.



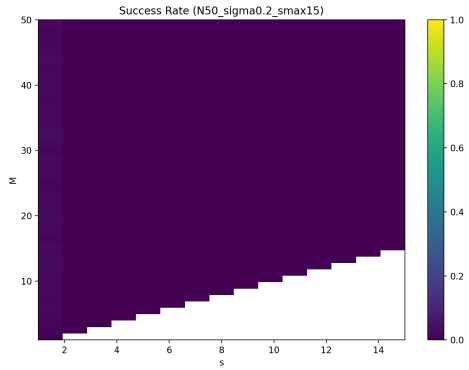
(a) $N = 20,\ \sigma = 0.05$



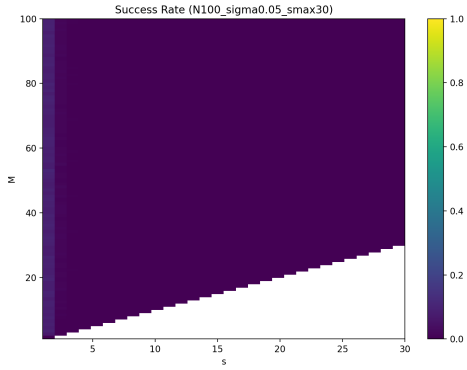
(b) $N = 20,\ \sigma = 0.2$



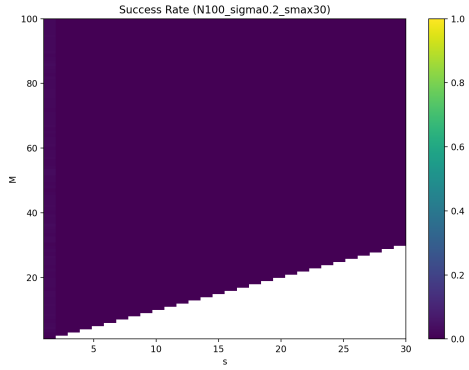
(c) $N = 50,\ \sigma = 0.05$



(d) $N = 50,\ \sigma = 0.2$

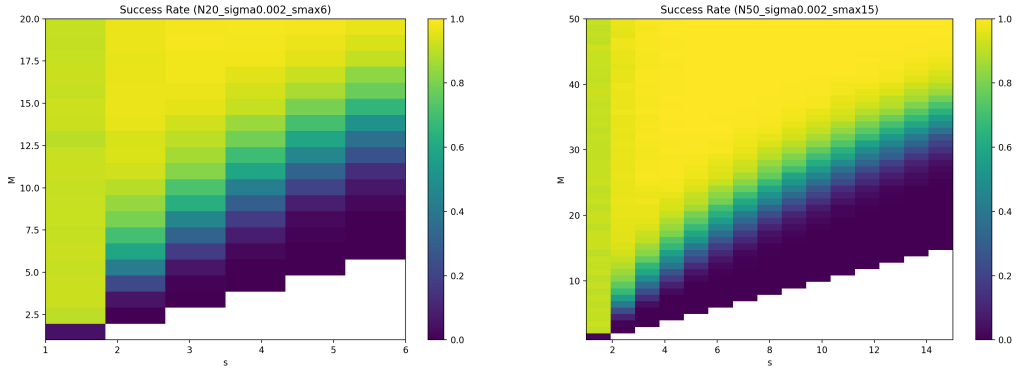


(e) $N = 100,\ \sigma = 0.05$

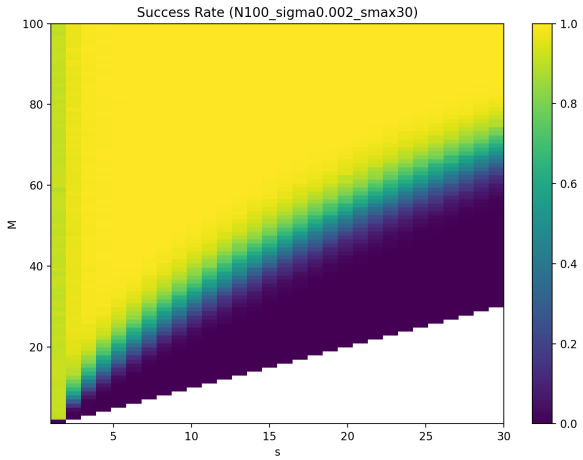


(f) $N = 100,\ \sigma = 0.2$

Figure 5: Noisy phase transition (known $\|n\|_2$ stopping): success rate for $N \in \{20, 50, 100\}$ and $\sigma \in \{0.05, 0.2\}$. Success is defined as Normalized Error $< 10^{-3}$.



(a) $N = 20,\ \sigma = 0.002$ (b) $N = 50,\ \sigma = 0.002$



(c) $N = 100,\ \sigma = 0.002$

Figure 6: Additional noisy phase transition results for the residual-based stopping rule (unknown s but known $\|n\|_2$) under a smaller noise level $\sigma = 0.002$. Success is defined as Normalized Error $< 10^{-3}$.

Here, I also tried $\sigma = 0.2$ with other parameters unchanged and the result is shown in Fig. 6. Similar to the known-s case, a clear transition boundary is observed at $\sigma = 0.002$. The residual-based stopping rule adapts the number of iterations to the noise floor, and it can yield slightly higher success rates near the transition boundary by avoiding unnecessary extra iterations that might select spurious atoms under Normalized Error $< 10^{-3}$. Therefore, under the condition of Normalized Error $< 10^{-3}$, OMP with known $\|n\|_2$ seems to be better than OMP based on known s near the transition boundary.

4 Decode a Compressed Message

(a) Guess the message from the compressed images

Fig. 7 shows the compressed measurements Y_1, Y_2, Y_3. They are all completely noise, so I can not guess the message from these images.

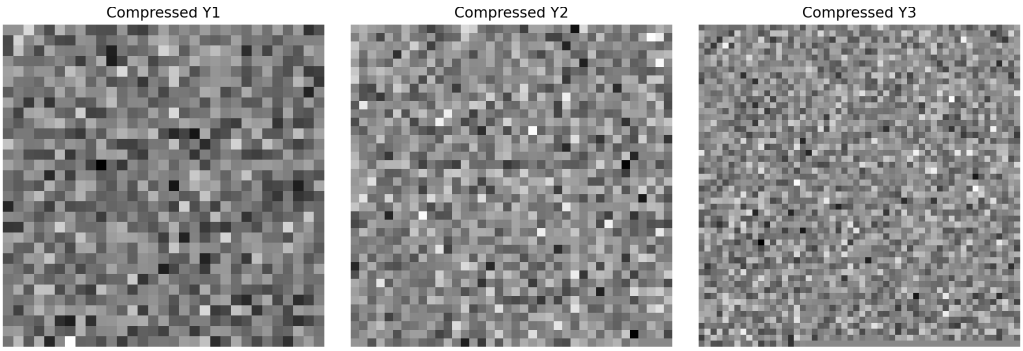
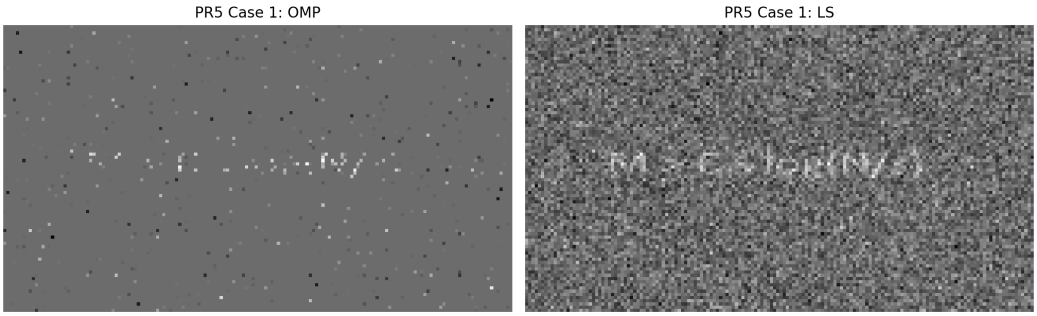


Figure 7: Compressed measurements Y_1, Y_2, Y_3 visualized as images.

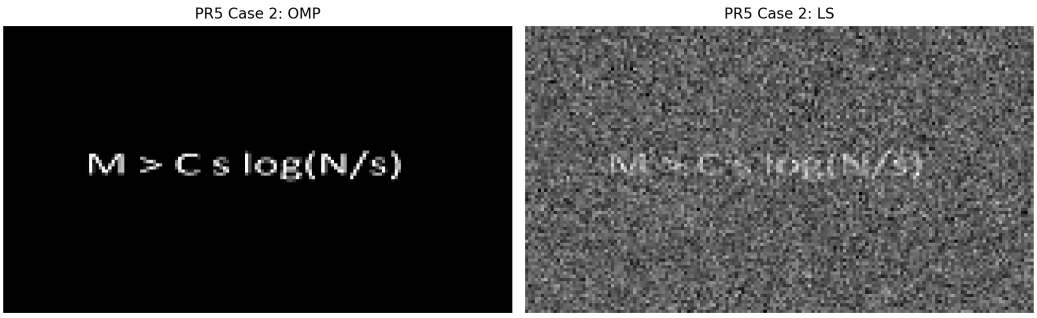
(b) Recover X using OMP and compare with LS (including stopping criteria)

I recover X from each measurement Y_i using OMP, reshape the estimate into a 90×160 image, and compare it with the Least Squares (LS) baseline. Since the sparsity level s is unknown in this task, I use a residual-based stopping rule: $\frac{\|y - Ax^{(k)}\|_2}{\|y\|_2} \leq \texttt{tol} \quad \text{or} \quad k \geq \texttt{smax}$. In my implementation, I set $\texttt{tol} = 10^{-2}$ and $\texttt{smax} = 600$, which provides a practical trade-off between reconstruction quality and runtime.

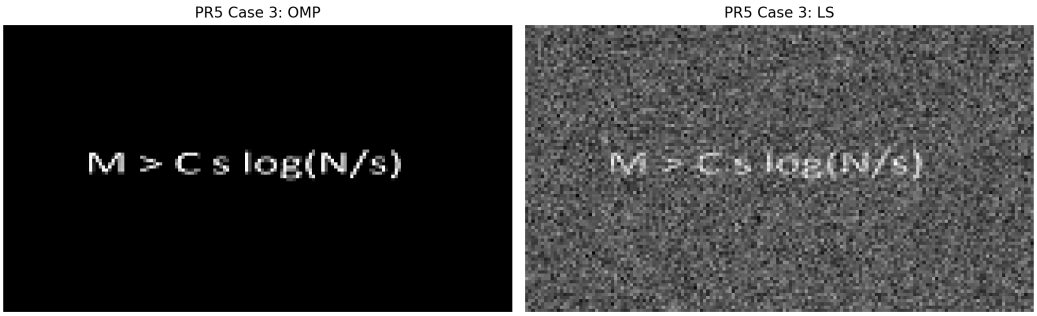
Fig. 8 shows that OMP reconstructs a readable message in Case 2 and Case 3, while LS produces a minimum-norm solution that does not promote sparsity and largely looks like noise yet still readable in Case 3.



(a) Case 1: OMP vs. LS.



(b) Case 2: OMP vs. LS.



(c) Case 3: OMP vs. LS.

Figure 8: Recovered images for the compressed message task.

(c) Which corrupted image gives the best OMP result and why

Case 3 gives the best OMP result (Case 2 is similarly good). Intuitively, this case provides more useful measurement information, so OMP can identify the correct support more reliably, leading to a sharper reconstruction. In contrast, the more corrupted case, Case 1, makes atom selection less reliable, so the recovered image is much less readable.

(d) Guess about the meaning of this message

The decoded message displays the formula:

$$M > Cs\log(N/s)$$

After I googled the result, I found out this is known as the "Fundamental Bound of Compressed Sensing" or the "Phase Transition Boundary". It mathematically states that to perfectly recover an s-sparse signal of ambient dimension N, the minimum number of measurements M must scale on the order of $s\log(N/s)$. This formula directly explains the sharp phase-transition behavior we plotted earlier: reliable recovery is only guaranteed once M crosses this specific theoretical threshold.

5 Decode a Compressed Audio Signal

5.1 (a) Guess the message by simply playing the compressed signal

The compressed signal y sounds noisy when played directly, so I can not guess the hidden message.

5.2 (b) Reconstruct and play $\hat{x} = D\hat{s}$ for $k \in K$, and compare OMP with LS

The signal model is $x = Ds$ where $s \in \mathbb{R}^{15980}$ is 100-sparse, and measurements satisfy $y = Ax = ADs$. For each $k \in K = \{10, 50, 100, 200, 300, 1000, 2000, 3000\}$, I form $y_k = y_{1:k}$ and $A_k = A_{1:k,:}$, reconstruct $\hat{s}$ from y_k, and synthesize $\hat{x} = D\hat{s}$.

From the reconstructed audio, the decoded spoken message is *"I love linear algebra."* In my experiments, LS requires a much larger number of measurements to become intelligible: the message only becomes audible around $k = 1000$, and the reconstructed audio still contains noticeable background noise. In contrast, OMP produces an intelligible reconstruction with far fewer measurements: the message is already clear around $k = 50$, with substantially less audible noise. As k increases (e.g., $k = 2000$ or $k = 3000$), the OMP reconstruction becomes nearly noise-free, while the LS reconstruction still contains residual noise.

Since the report is submitted as a PDF, the reconstructed `.wav` files for both OMP and LS at the tested k values are included in the project submission zip.

5.3 (c) Minimum number of measurements $K_{\min}$ and what it depends on

Based on perceptual quality, the minimum number of measurements sufficient to understand the message is $K_{\min} = 50$ for OMP, while for LS it is much larger, around $K_{\min} = 1000$ in my tests.

This threshold $K_{\min}$ depends on several practical factors. A higher noise level generally requires more measurements to make the reconstructed speech intelligible. It also depends on how complex the underlying signal is, the sensing matrix A, the reconstruction algorithm , and the chosen stopping rule.

The relation to the size and sparsity of s is intuitive. The length of s (ambient dimension N) indicates how many unknown coefficients could potentially be present: a larger N typically makes the problem harder and increases the needed measurements. The sparsity level (number of nonzeros) controls how much "essential information" is in s: the sparser the vector (fewer nonzeros), the fewer measurements are needed; if s is less sparse (more nonzeros), more measurements are required.

Measurements beyond $K_{\min}$ improve perceptual quality even after the message becomes understandable. In particular, increasing k typically reduces reconstruction error and audible artifacts, leading to cleaner speech, less background noise, and more stable reconstructions across trials.

References

[1] Joel A. Tropp. Greed is Good: Algorithmic Results for Sparse Approximation. *IEEE Transactions on Information Theory*, 50(10):2231–2242, 2004.